

1. User Management & Email Notification – Frontend (FE) & Backend (BE)

Overview:

The User Management module is the foundation of the system. It controls how users **Register**, **Log-in**, and **Access** their specific roles within the platform.

Each type of **Tourist**, **Agent**, and **Administrator** interacts with the system through role-based permissions and functionalities.

This module integrates a **Secure Payment Gateway** to manage user payments, billing details, and transaction records, ensuring reliable and encrypted financial operations. Additionally, it includes an **Email Notification System** to deliver timely confirmations, security alerts, and account-related updates such as registration verification, password resets, and payment confirmations.

Frontend (FE):

- Provides interactive forms for **User Registration** and **Login**, with instant validation for input accuracy.
- Offers **Role Selection Options** during signup (User, Agent, Admin).
- Supports **Password Reset**, **Email Verification**, and **Multi-language (Sinhala, Tamil, English)** accessibility.
- **Update & Modify Identification Details.**
- Enables **Secure Payment Gateway** integration for transactions.
- **Displays confirmation** messages, downloadable receipts, and billing invoices after successful payments.
- **Shows email notification** pop-ups or banners for actions that require user confirmation (ex: "Check your email for verification link").

Backend (BE):

- Handles **Secure Authentication & Authorization** using role-based access control (RBAC).
- Encrypts passwords and user data to ensure privacy.
- Manages **User Roles, Permissions, & Account Status** (active, pending, deactivated).
- **Stores & Manages Payment-related Data** such as transaction logs, invoices, and billing history.
- **Validates** payments, **updates** user payment status, and **triggers** confirmation workflows.
- **Provides APIs** for sending automated email notifications (account creation, password reset, booking confirmation, profile update alerts).
- **Manages communication logs** and notifications (emails/SMS/chatbot responses).
- Ensures **Secure & Reliable Email Delivery** using SMTP or third-party email services.

- Integrates **session and token-based authentication (JWT)** for secure access.

Purpose:

The purpose of this module is to ensure **Secure, Role-based** access to the platform while enabling users to **Manage** and **Personalize** their accounts efficiently. It also facilitates **Secure Financial Transactions** through an integrated payment gateway, allowing the system to handle payments, billing information, and transaction records in a reliable and transparent manner.

By incorporating an **Email Notification System**, the module keeps users informed through timely confirmations and alerts related to account activities, security actions, and payment events. Together, these capabilities enhance user trust, ensure data security, support smooth financial interactions, and provide a dependable foundation for all platform operations.

2. Agent Portal – Frontend (FE) & Backend (BE)

Overview:

The **Agent Registration module** enables travel agents or independent drivers to join the platform and list their travel-related services. This scope ensures that travellers can access verified transportation options while agents can manage their business profiles, vehicles, and booking schedules efficiently.

Frontend (FE):

- Allows agents to **Add Vehicles, & Driver** with details.
- Includes an **Interactive Dashboard** for agents to **Manage** their registered **Vehicles** and **Update** availability status.
- Displays **Booking Requests & Assigned Trips** for agents with details such as customer name, destination, and dates.
- Provides options to **Accept or Decline Booking Requests**, **Update** trip **Status**, and mark trips as completed.
- Allows **Profile Updates** and **Performance Tracking** (ratings, reviews, and trip statistics).
- Shows **Performance Analytics** such as ratings, reviews, completed trips, cancellation history, and revenue reports.

Backend (BE):

- **Stores Agent/Company/Vehicle/Driver details** in the database.
- Allows **Profile Updates & Provides** a **Page** for Tourists to view Agent details.
- **Links Vehicles, & Driver to Packages** or direct bookings made by tourists.
- Handles **Availability Management**, **Trip Allocation**, and **Real-Time Updates** on booking status.
- Provides **RESTful APIs** for CRUD (Create, Read, Update, Delete) operations on vehicles, bookings, and agent data.
- Stores and manages **Ratings & Reviews**, calculates performance scores, and provides data for analytics dashboards.
- **Maintains Real-Time Booking Updates**, ensuring tourists see accurate availability.

Purpose:

The purpose of the **Agent Portal** is to provide travel agents and drivers with a dedicated platform to register their services and manage daily operations efficiently. It enables agents to list and maintain their vehicles and driver details, update availability, and handle booking requests in an organized manner.

Through **Performance Tracking** features such as ratings, reviews, trip history, and revenue reports, agents can monitor service quality and improve customer satisfaction. At the same time, the portal ensures that tourists have access to verified, reliable, and well-managed transportation services.

3. Chatbot & Hotel Suggestion Module– Frontend (FE) & Backend (BE)

Overview:

The **Chatbot & Hotel Suggestion module** enhances user interaction by providing instant assistance and intelligent hotel recommendations. The chatbot helps users by answering common questions, guiding them through the system, and assisting with navigation.

This module also includes a **Hotel Information and Suggestion System**, where hotels are stored in a database and suggested to users based on the selected district or location. The module does **not handle hotel booking** but focuses on **displaying verified hotel details and reviews** to help users make informed travel decisions.

Frontend (FE):

- Provides an **integrated chatbot** widget accessible from all major pages in the system.
- Allows users to **ask questions and receive instant automated responses**.
- Displays typing indicators, chat history, and suggested replies for better interaction.
- **Shows hotel suggestions based on the district or location selected by the user.**
- **Displays Hotel Details** such as name, location, images, description, facilities, and star rating.
- Allows tourists to **Add Reviews** and **Ratings** for Hotels.
- **Displays Hotel Reviews** and **Ratings** submitted by previous users.

Backend (BE):

- Integrates with an **AI-based Chatbot Engine** to process user queries.
- **Stores chatbot conversation** logs for quality improvement and admin review.
- Handles chatbot fallback responses and routes unresolved questions to human support if needed.
- **Maintains message logs**, and **user communication** settings.
- **Provides APIs to Retrieve Hotels** based on district, location.
- **Stores Hotel Information** securely in the database (hotel name, district, address, facilities, images, ratings).
- **Stores and Retrieves Hotel Reviews** from the database.
- **Calculates** and maintains average hotel **Ratings Dynamically**.

Purpose:

The purpose of this module is to **Improve User Assistance & Decision-Making** by combining intelligent chatbot support with location-based hotel suggestions.

By maintaining a dedicated **Hotel Information Database** and integrating **Hotel Reviews**, users can explore suitable accommodation options in a selected area without directly performing bookings. This approach enhances transparency, builds user trust, reduces confusion during trip planning, and improves overall platform usability through guided assistance and informed hotel discovery.

4. User & Admin Portal – Frontend (FE) & Backend (BE)

Overview:

The **User & Admin Portal** module serves as a centralized dashboard for both **Tourists & Administrators**. For tourists, it provides a personalized space to **Manage Trips, Bookings, & Feedback**. For administrators, it acts as a **Master Control Panel** to monitor platform activity, verify users, and enforce system policies.

This module ensures smooth coordination between users, agents, hotels, and packages while giving **Administrators full visibility and control** over platform operations.

Frontend (FE):

- Displays a **Customized Dashboard** showing ongoing and completed trips.
- Allows users to **View Booking Details**, including vehicle information, driver details, travel route, schedule, price breakdown, and status updates.
- Includes **Options to Download** invoices, receipts, and travel documents.
- Provides an **Admin Master Dashboard** displaying system statistics (users, agents, hotels, packages, bookings).
- Allows admins to **Approve or Reject Agent, Hotel** registrations after verification.
- Enables admins to **Enable, Disable, or Remove Travel Packages** that violate platform policies.

Backend (BE):

- **Fetches & Updates** user-specific booking, and itinerary data.
- **Update & Modify Identification Details**
- Manages **Package Activation, Deactivation, & Removal** processes (Admin).
- Handles **Verification** workflows for **Agent** and **Hotel** registration approvals
- Provides **Secure APIs** for managing trip data and user dashboard updates.
- Ensures all **User Data Flows Smoothly** between modules using efficient synchronization processes.
- **Generates & Stores Invoices**, receipts, and travel documents in digital format (PDF or downloadable files).
- Integrates data synchronization between modules (Package, Hotel, Agent).

Purpose:

The purpose of this portal is to help **Tourists Easily Manage Their Travel Experience** in one place. It allows them to view ongoing and completed trips, check full booking details, and download invoices, receipts, or travel documents whenever needed. This makes trip planning and tracking simple, clear, and convenient.

The purpose of this portal is to give **Administrators a Centralized Dashboard** to monitor and control platform activities. It helps admins oversee user trips, verify system data, and ensure smooth platform operations. By providing clear visibility and management tools, the portal supports better decision-making and effective system governance.

5. Package Definition – Frontend (FE) & Backend (BE)

Overview:

The Package Definition module is responsible for **Creating, Managing, and Presenting** travel packages within the system. It allows **Agents** to define **Destinations, Activities, Prices, and Schedules** that form complete travel experiences.

This module is enhanced by integrating a **Festival Calendar, Package Search and Filtering** enabling tourists to plan trips around Sri Lanka's cultural, religious, and seasonal events.

Additionally, the module supports a **Package Review System**, allowing tourists to share feedback and ratings about Packages.

Frontend (FE):

- Provides a **form-based interface** for Agent to create or edit packages.
- Includes **Input fields** for **Destinations, Duration, Daily Activities, and Package Type** (Standard/Custom/Solo).
- **Displays Packages** to users with images, itineraries, reviews, and total cost breakdowns.
- Offers **Search and Filter Tools** by location, budget, season, and category (eco, cultural, adventure).
- **Highlights packages associated with major festivals** (e.g., Kandy Esala Perahera, Vesak, Sinhala & Tamil New Year).
- Allows tourists to **Add Reviews and Ratings** for packages they have booked.

Backend (BE):

- Handles **CRUD operations** for all **Package** and destination data.
- Links packages with other entities such as hotels, vehicles, and festivals through relational data models.
- Manages **Real-Time Pricing Updates, Availability, and seasonal adjustments**.
- Provides **APIs for package discovery, search filters, and detailed viewing**.
- Integrates reviews and ratings for each package.
- Enables **admin control** to activate, deactivate, or modify travel packages.
- **Stores and manages festival and event data** linked to destinations and travel date.
- **Stores, Retrieves, and Manages** package **Reviews and Ratings** in the database.

Purpose:

The purpose of this module is to offer well-structured, **Customizable** travel packages while enhancing trip planning through festival awareness and user feedback. By integrating a **Review and Rating system**, tourists can share their travel experiences, while future users gain trustworthy insights into package quality. This improves transparency, increases user confidence, supports continuous service improvement, and enhances overall engagement with the platform. Together with festival-based planning, this module promotes culturally meaningful, well-timed, and user-informed travel experiences across Sri Lanka.